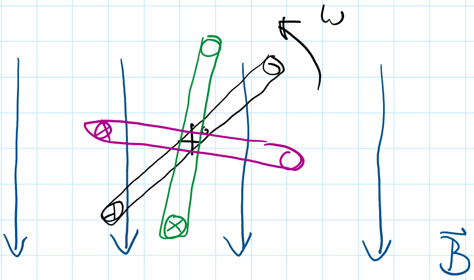
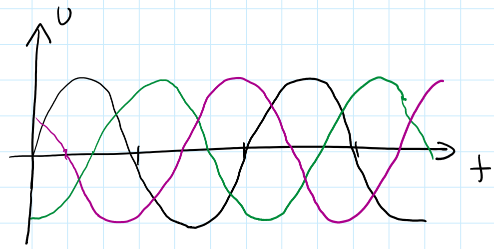
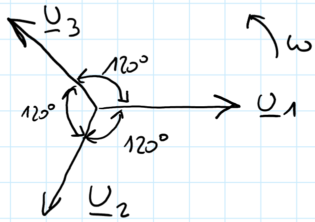
MEHRPHASENSYSTEME

- $\hat{U}$ ist immer gleich
- ω — " —
- Spezialfall Drehstrom = 3 Phasen
  - Spannungen mit Phasenverschiebung 120°
  - $\underline{U}_1 = U\angle 0°$ $\underline{U}_2 = U\angle -120°$ $\underline{U}_3 = U\angle -240° =$ $= U\angle +120°$







- Strang = 1 Energiequelle mit Hin- und Rückleitung zum Verbraucher

STERNSCHALTUNG $\curlywedge$

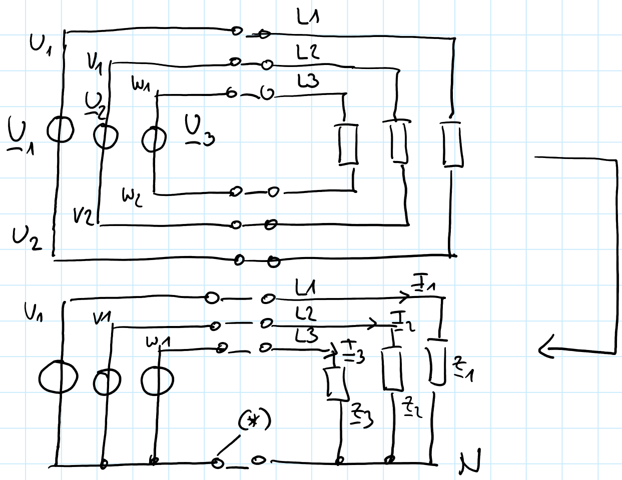
- N... Neutralleiter
- Vier- oder Dreileiter-System (*)
- Sternspannung = Spannung Außenpunkt – Sternpunkt
- Strangspannung = Spannung an Wicklung

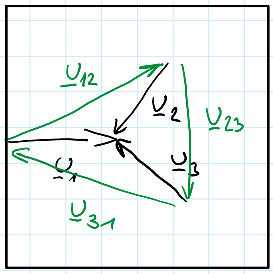
$U_1 = U_2 = U_3 = U_\curlywedge$

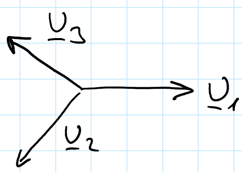
$\underline{U}_{12} = \underline{U}_1 - \underline{U}_2$

$U_{12} = U_{23} = U_{13} = U$

$U = 2\, U_\curlywedge \cos 30° = \sqrt{3}\, U_\curlywedge$





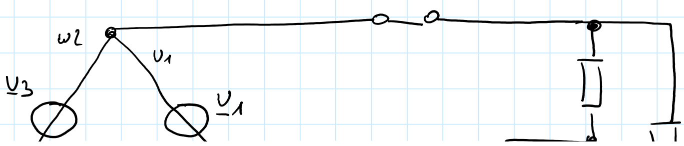


NS: $U_\curlywedge = 230\,V$ $U = 400\,V$

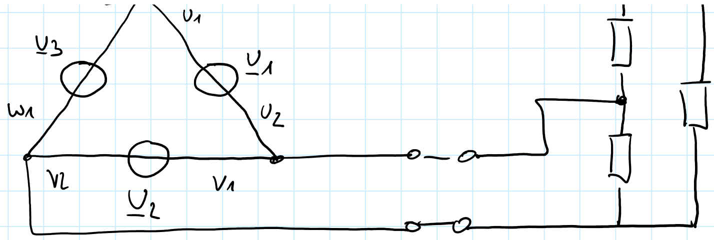
380 kV
110 kV } i.d.R. Außenleiter + Effektivwert
20 kV

DREIECKSCHALTUNG $\triangle$

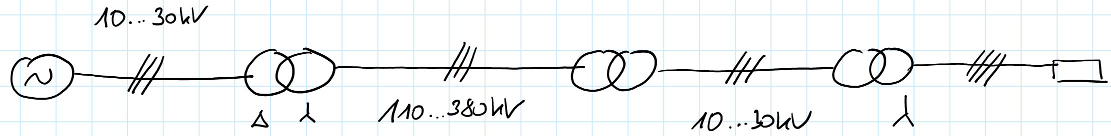
- Dreileitersystem
- Strangspannung = Außenleiter-

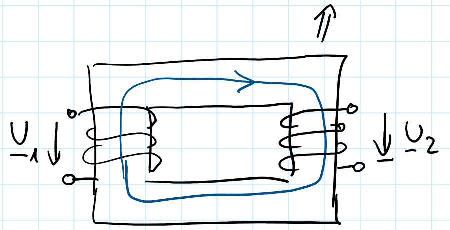


- Strangspannung = Außenleiterspannung
- $U_{St} = U = U_\Delta$
- künstlicher Sternpunkt mit 3 gleichen Widerständen



SPANNUNGSVERSORGUNG





TRANSFORMATOR

VERBRAUCHERSCHALTUNG IM ⅄

- symmetrische Last $\underline{Z}_1 = \underline{Z}_2 = \underline{Z}_3$
- $\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0 = \underline{I}_N$
- $I_1 = I_2 = I_3 = I_\lambda = \frac{U_\lambda}{Z}$
- Leistung ?

$\varphi_{U1,U2,U3} = 0, -120°, 120°$

$$p(t) = U \cdot I \cdot \cos\varphi + UI\cos(2\omega t + \varphi_u + \varphi_i) \qquad \varphi = \varphi_U - \varphi_i$$

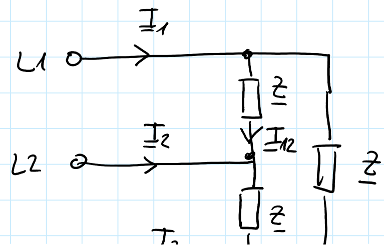
$$\left.\begin{aligned} p(t) = {} & U_\lambda I_\lambda \cdot \cos\varphi - U_\lambda I_\lambda \cos(2\omega t + \varphi) \\ & + U_\lambda I_\lambda \cos\varphi - U_\lambda I_\lambda \cos(2\omega t + \varphi - 240°) \\ & + U_\lambda I_\lambda \cos\varphi - U_\lambda I_\lambda \cos(2\omega t + \varphi + 240°) \end{aligned}\right\} \hat{=} 0$$

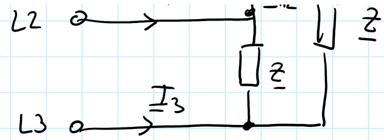
$$P(t) = 3\, U_\lambda I_\lambda \cos\varphi + 0 = \text{const.}$$

$$\underline{S} = 3\, U_\lambda I_\lambda \angle\varphi = \sqrt{3}\, U I \angle\varphi = P + jQ$$

VERBRAUCHERSCHALTUNG IM △

- $U_\Delta = U = \sqrt{3}\, U_\lambda$
- $I_\Delta = I_{12} = \frac{U_\Delta}{Z}$
- $I = I_1 = I_2 = I_3 = \sqrt{3}\, I_\Delta$



- $\underline{I} = \underline{I}_1 = \underline{I}_2 = \underline{I}_3 = \sqrt{3}\ \underline{I}_\Delta$ 
- $p(t) = 3\, U_\Delta I_\Delta \cos\varphi$ = const. !
- $S_\Delta = 3\ S_\lambda$ bei Z = const.
- $\underline{S} = 3\ U_\Delta I_\Delta \angle\varphi$

UNSYMMETRISCHE BELASTUNG

- Starres Netz = ideale Spannungsquelle + minimale Leistungsverluste
- Sternpunktleiter belastbar

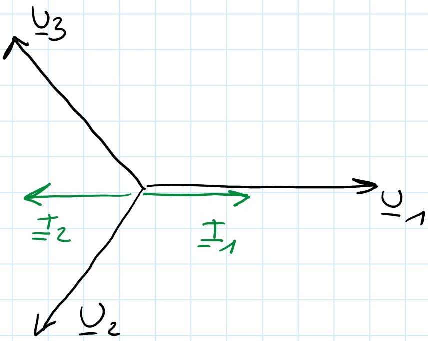
BSP VIERLEITERNETZ UNSYMMETRISCH L3: unbelastet

geg. $U = 400V$ L1: $R = 23\,\Omega$ L2: $I = 10A$ $\cos\varphi = 0{,}5$ $(\varphi = 60°)$

$\underline{U}_1 = 230V\angle 0°$ $\underline{I}_1 = 10A\angle 0°$

$\underline{U}_2 = 230V\angle -120°$ $\underline{I}_2 = 10A\angle -120° - 60°$

$\underline{\underline{I_N}} = \underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 10A\angle 0° + 10A\angle -180° = \underline{0A}$



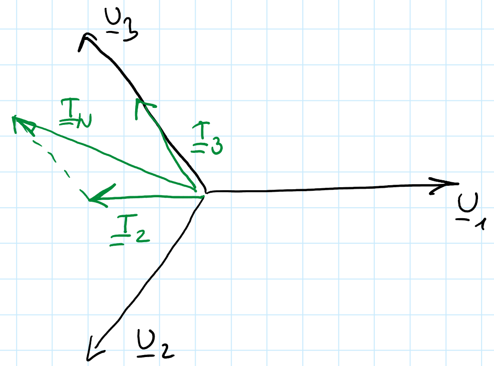
Annahme Anschluss R an L3 (statt L1)

$\underline{U}_3 = 230V\angle +120°$ $\underline{I}_3 = 10A\angle +120°$

$\underline{I}_N = \underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0 + 10A\angle -180° + 10A\angle +120° =$

$= 10A(-1 + j\varnothing) + 10A(-0{,}5 + j\,0{,}866) =$

$= 10A(-1{,}5 + j0{,}866) = 17{,}32A\angle +150°$

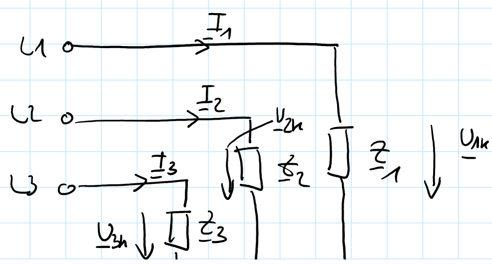


SPEZIALFALL DREILEITERNETZ STERNSCHALTUNG $\curlywedge$

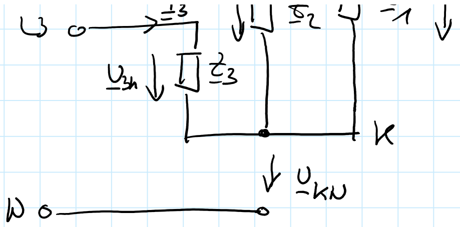
$\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0$

$\underline{U}_{1k} \cdot \underline{Y}_1 + \underline{U}_{2k} \cdot \underline{Y}_2 + \underline{U}_{3k}\, \underline{Y}_3 = 0$

$(\underline{U}_{1N} - \underline{U}_{kN})\underline{Y}_1 + (\underline{U}_{2N} - \underline{U}_{kN})\underline{Y}_2 + (\underline{U}_{3N} - \underline{U}_{kN})\underline{Y}_3 = 0$



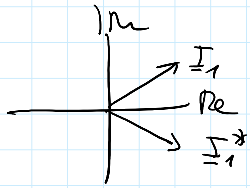
$$(\underline{U}_{1W}-\underline{U}_{KW})\underline{Y}_1+(\underline{U}_{2W}-\underline{U}_{KW})\underline{Y}_2+(\underline{U}_{3W}-\underline{U}_{KW})\underline{Y}_3=0$$



$$\underline{U}_{1W}\underline{Y}_1+\underline{U}_{2W}\underline{Y}_2+\underline{U}_{3W}\underline{Y}_3=\underline{U}_{KW}(\underline{Y}_1+\underline{Y}_2+\underline{Y}_3)$$

$$\underline{U}_{KW}=\frac{\underline{U}_{1W}\cdot\underline{Y}_1+\underline{U}_{2W}\cdot\underline{Y}_2+\underline{U}_{3W}\,\underline{Y}_3}{\underline{Y}_1+\underline{Y}_2+\underline{Y}_3}$$

Sternpunktspannung



$$\underline{S}=\underline{U}_{1K}\underline{I}_1^*+\underline{U}_{2K}\cdot\underline{I}_2^*+\underline{U}_{3K}\underline{I}_3^*=$$

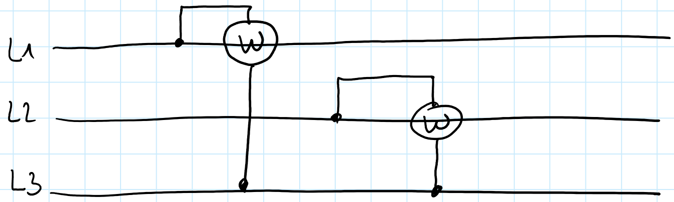
$\underline{U}_{1K}=\underline{U}_{1W}-\underline{U}_{KW}$

$$=\underline{U}_{1W}\underline{I}_1^*+\underline{U}_{2W}\underline{I}_2^*+\underline{U}_{3W}\underline{I}_3^*-\underline{U}_{KW}\underbrace{(\underline{I}_1^*+\underline{I}_2^*+\underline{I}_3^*)}_{=0}$$

$\underline{I}_3^*=-\underline{I}_1^*-\underline{I}_2^*$

$$=(\underline{U}_{1W}-\underline{U}_{3W})\underline{I}_1^*+(\underline{U}_{2W}-\underline{U}_{3W})\underline{I}_2^*=$$

$$=\underline{U}_{13}\underline{I}_1^*+\underline{U}_{23}\underline{I}_2^*$$



ARON-SCHALTUNG

2 Wattmeter

Unsymmetrische Belastung in Δ

- SUMME AUSSENLEITERSTRÖME $=0$
- AUSSENLEITERSTRÖME NICHT SYMMETRISCH
- $\underline{S}=\underline{U}_{1W}\underline{I}_1^*+\underline{U}_{2W}\underline{I}_2^*+\underline{U}_{3W}\underline{I}_3^*$